UNIT 3 · TOPIC 12(B)

Electric Motors and Their Characteristics

Syllabus topic: Measurement of engine power; electric motors and their characteristics including batteries

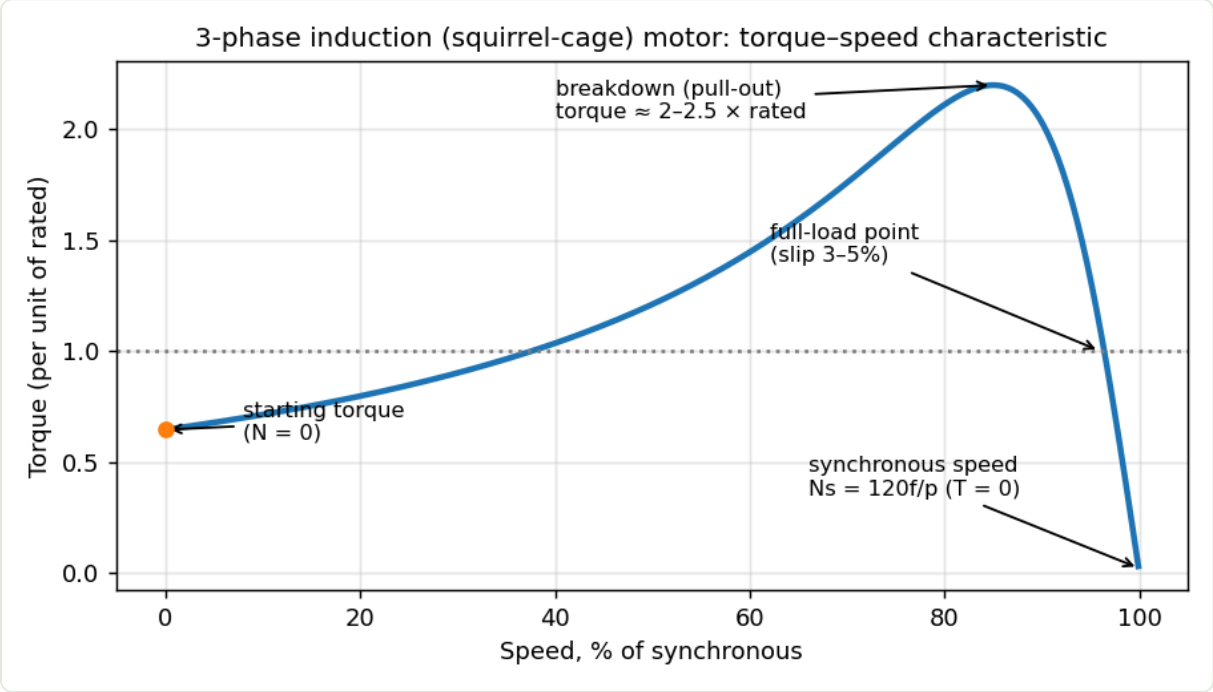
Types

Motor	Use
Three-phase SQUIRREL-CAGE INDUCTION — simple, rugged, cheap, minimal maintenance	The farm standard for pumps, chaff cutters, threshers and mills where a 3-φ supply exists
Single-phase — split-phase; CAPACITOR-START (high starting torque); capacitor-run	Small machines on domestic supply
DC/BLDC	Battery-operated equipment, drones, e-tractors

Induction motor characteristics

$$\text{Synchronous speed } N_s = 120f/p;\ \text{slip } s = (N_s - N)/N_s$$

- 50 Hz, 4-pole → Ns = **1500 rpm**. Full-load slip **3–5%** (N ≈ 1440 rpm).
- Torque rises to the **breakdown (pull-out) torque** (≈ **2–2.5× rated**), then falls to zero at Ns.
- Efficiency **85–92%**; power factor **0.8–0.9** at full load (poor at light load).
- Overload capacity is limited — size the motor to the peak demand (chaff-cutter flywheels help).



3-phase induction (squirrel-cage) motor — torque–speed characteristic

Measuring motor power

$$\text{Input (two-wattmeter method, 3-}\varphi\text{): } P = \sqrt{3}\cdot V_L \cdot I_L \cdot \cos\varphi;\ \text{efficiency} = \text{output/input}$$

Measure the output with a brake/dynamometer.

Motors vs engines

Motors offer **85–90%** efficiency, no idle fuel, low noise, less maintenance and an instant start — but they are tied to the grid.

Practice MCQs

Q157. The standard electric motor for farm use (3-φ available) is:

(a) DC series
(b) Squirrel-cage induction
(c) Synchronous
(d) Universal

Q158. Synchronous speed of a 4-pole, 50-Hz motor is:

(a) 750
(b) 1000
(c) 1500
(d) 3000 rpm

Q159. If it runs at 1440 rpm, slip =

(a) 2%
(b) 4%
(c) 6%
(d) 8%

Q160. Full-load slip of induction motors is typically:

(a) 0%
(b) 3–5%
(c) 15–20%
(d) 40%

Q161. Breakdown (pull-out) torque of an induction motor is about:

(a) Equal to rated
(b) 2–2.5× rated
(c) 10× rated
(d) 0.5× rated

Q162. For high starting torque on single-phase supply, use a:

(a) Plain split-phase motor
(b) Capacitor-start motor
(c) Shaded pole
(d) Synchronous motor

Q163. Three-phase motor input power is measured by:

(a) One ammeter
(b) Two-wattmeter method
(c) Hydrometer
(d) Tachometer only

Q164. $P = \sqrt{3} \cdot V \cdot I \cdot \cos\varphi$; cosφ here is the:

(a) Slip
(b) Power factor
(c) Efficiency
(d) Torque ratio

Q165. Efficiency of well-loaded induction motors is about:

(a) 40–50%
(b) 60–70%
(c) 85–92%
(d) 100%

Answers

Q157 — (b) Squirrel-cage induction
Q158 — (c) 1500 — *Ns = 120×50/4 = 1500 rpm.*
Q159 — (b) 4% — *s = (1500-1440)/1500 = 4%.*
Q160 — (b) 3–5%
Q161 — (b) 2–2.5× rated
Q162 — (b) Capacitor-start motor
Q163 — (b) Two-wattmeter method
Q164 — (b) Power factor

Q165 — (c) 85–92%

Sources: Tractors and their Power Units (Liljedahl et al.); Kepner et al.; BIS publications; ISO/OECD/ASABE documentation; NPTEL lectures on farm power and IC engines.
